Abhishek Kumar

Senior Applied Scientist, AWS AI | New York, NY

iyamabhishek09@gmail.com +1 564-218-2568

Google Scholar in/iyamabhishek iyamabhishek.github.io

Research Interests

LLM agents and agent evaluation; retrieval-augmented generation, retrieval and memory; reasoning and planning; multi-turn and multi-session dialogue; LLM post-training and reinforcement learning.

Google Scholar (Aug 2026): 381 citations, h-index 9, i10-index 9.

Education

University of Maryland, College Park 2019 – 2021
M.S. in Computer Science (GPA 3.94/4.0) *College Park, MD*

Indian Institute of Technology (IIT) Patna 2014 – 2018
B.Tech. in Computer Science and Engineering (GPA 9.13/10.0) *Patna, India*

- Best B.Tech. Thesis Award; undergraduate researcher at the AI-NLP-ML Lab.

Experience

Amazon Web Services (AWS AI) New York, NY
Senior Applied Scientist (May 2026–present); Applied Scientist II (Oct 2023–Apr 2026)

- **Agent-evaluation model (LLM-as-a-Judge).** Led the post-training — supervised fine-tuning followed by GRPO/RLVR reinforcement learning — of a large language model that automatically scores the step-by-step execution traces of AI agents across ∼20 categories of failure. Built the accompanying synthetic-data pipeline spanning 149 agents and 6 agent frameworks to produce 1B+ training tokens.
- **Tool-specification quality & model distillation.** Owned end-to-end a system that audits and rewrites the tool and skill specifications that agents depend on to act correctly: a perturbation framework, a ∼60k-sample evaluation dataset, and an LLM-based quality-scoring and rewriting pipeline. Distilled compact 4B–14B models from frontier LLMs that outperformed the teacher model on specification linting.
- **Agent observability & user simulation.** Developed a graph-based root-cause-analysis method whose trace-pruning algorithm isolates the causally relevant subgraph of an agent's reasoning to explain why it failed. Built a persona-conditioned "user simulator" agent that carries out goal-oriented, tool-using multi-turn conversations to evaluate other agents at scale.
- **Open-source Strands SDKs.** Core science contributor to the Strands Agents SDK (a framework for building LLM agents, open-sourced in 2025); designed and validated the ReAct-style tool-use and reasoning experiments that became its foundational architecture. Open-sourced the user simulator (*ActorSimulator*) and the root-cause analyzer in the Strands Evals SDK. Filed 2 U.S. patents on agent observability and simulation.
- **Multi-agent Amazon Q Console.** Re-architected Amazon's enterprise cloud-assistant from a single-intent system into a multi-agent, tool-using orchestration; led a 3-scientist team on personalization and multi-turn evaluation, improving external customer satisfaction from ∼40% to ∼70%.
- **Agentic network diagnostics & retrieval.** Rebuilt AWS's VPC Reachability Analyzer assistant from a brittle static-prompt design into an agentic tool-use system (user satisfaction $<20\% \rightarrow >55\%$), and designed a FAISS embedding-based top-k retrieval shortlister that cut LLM latency 37% and enabled the launch of 3 assistant extensions.

Amazon (AGI / Alexa AI) Seattle, WA
Applied Scientist II *Jun 2021 – Sep 2023*

- **LLM post-training at ∼100B-parameter scale.** Authored a supervised fine-tuning / instruction-tuning runbook adopted by 20+ scientists, and built a data curation and filtering pipeline for model release that reduced scientist effort by ∼15%.
- **Model reliability & entity resolution.** Corrected 134k inconsistent training annotations, reducing model quality-gate failures by 60% (+166k successful dialogs/month); built an XGBoost re-ranker (+20% recall, +400k confirmations/month) with automated hyper-parameter optimization across 6 locales.

Alexa AI, Amazon Sunnyvale, CA
Applied Scientist Intern *May – Aug 2020*

- Built multi-session dialogue systems in PyTorch, using a Graph Attention Network to encode conversational context across sessions (+4% over baseline).

Prior Research Experience

- **Researcher**, Indian Institute of Science (IISc), Bangalore 2018–2019
  Temporal process-graph construction from scientific literature via information extraction.
- **Research Intern**, Kyoto University (with Prof. Sadao Kurohashi) 2017–2018
  Knowledge-enriched two-layered attention networks for sentiment analysis (NAACL 2018).
- **Research Intern**, Université Paul Sabatier, Toulouse 2018
  Semi-supervised hate-speech detection for French.
- **Research Intern**, University of Hamburg (DAAD WISE Fellow) 2017
  Aspect-based sentiment analysis; GermEval 2017 shared task (open-sourced LT-ABSA).

Publications

Name underlined; ∗ denotes equal contribution. Full list on Google Scholar.

[1] Hieu Man, Ro-ee Tal, Abhishek Kumar, JJ Cho, Benjamin Hsu. *Reasoning with Memory: Adaptive Information Management for Retrieval-Augmented Generation*. **ACL** 2026.

[2] Mintong Kang, Vinayshekhar Bannihatti Kumar, Shamik Roy, Abhishek Kumar, Sopan Khosla, Balakrishnan Murali Narayanaswamy, Rashmi Gangadharaiah. *FairGen: Controlling Sensitive Attributes for Fair Generations in Diffusion Models via Adaptive Latent Guidance*. **EMNLP** 2025.

[3] Abhishek Kumar, Trisha Mittal, Dinesh Manocha. *MCQA: Multimodal Co-attention Based Network for Question Answering*. arXiv, 2020.

[4] Abhishek Kumar, Asif Ekbal, Daisuke Kawahara, Sadao Kurohashi. *Emotion helps Sentiment: A Multi-task Model for Sentiment and Emotion Analysis*. **IJCNN** 2019.

[5] Md Shad Akhtar, Abhishek Kumar, Asif Ekbal, Chris Biemann, Pushpak Bhattacharyya. *Language-Agnostic Model for Aspect-Based Sentiment Analysis*. **IWCS** 2019.

[6] Abhishek Kumar, Daisuke Kawahara, Sadao Kurohashi. *Knowledge-enriched Two-layered Attention Network for Sentiment Analysis*. **NAACL** 2018.

[7] Md Shad Akhtar*, Abhishek Kumar*, Deepanway Ghosal*, Asif Ekbal, Pushpak Bhattacharyya. *A Multilayer Perceptron based Ensemble Technique for Fine-grained Financial Sentiment Analysis*. **EMNLP** 2017.

[8] Abhishek Kumar, Abhishek Sethi, Md Shad Akhtar, Asif Ekbal, Chris Biemann, Pushpak Bhattacharyya. *IITPB at SemEval-2017 Task 5: Sentiment Prediction in Financial Text*. **SemEval** 2017.

[9] Eugen Ruppert, Abhishek Kumar, Chris Biemann. *LT-ABSA: An Extensible Open-Source System for Document-Level and Aspect-Based Sentiment Analysis*. **GSCL** 2017.

Patents

[1] Abhishek Kumar, et al. *AgentOps: Intelligent Agent-Based Observability System for Generative AI Applications*. U.S. Patent (filed 2025).

[2] Abhishek Kumar, et al. *Managed Agentic System Simulation and Test*. U.S. Patent (filed 2025).

Honors & Awards

- Best B.Tech. Thesis Award, Dept. of CSE, IIT Patna (2018).
- DAAD WISE Fellowship (2017) — research internship, University of Hamburg, Germany.
- Honda Research Institute Grant (2017) — research internship, Kyoto University, Japan.
- Conference travel grants: Kyoto University (NAACL 2018), IIT Patna (ACL 2017).
- Coal India Limited Scholarship — outstanding academic record, five consecutive years.

Service, Mentorship & Open Source

- **Area Chair:** *Who Verifies the Agents?* Workshop, NeurIPS 2026.
- **Reviewer:** ACL 2026, EMNLP 2021 & 2026, NeurIPS 2026, ACM CHI 2026.
- **Open source:** core contributor to the Strands Agents SDK and Strands Evals SDK; open-sourced ActorSimulator (agentic user simulator) and a graph-based Root-Cause Analyzer.
- **Mentorship:** mentored three research interns (2022, 2024, 2025); all received return offers, and the 2025 intern's work was submitted to ACL ARR.